

Week 5 - part 5

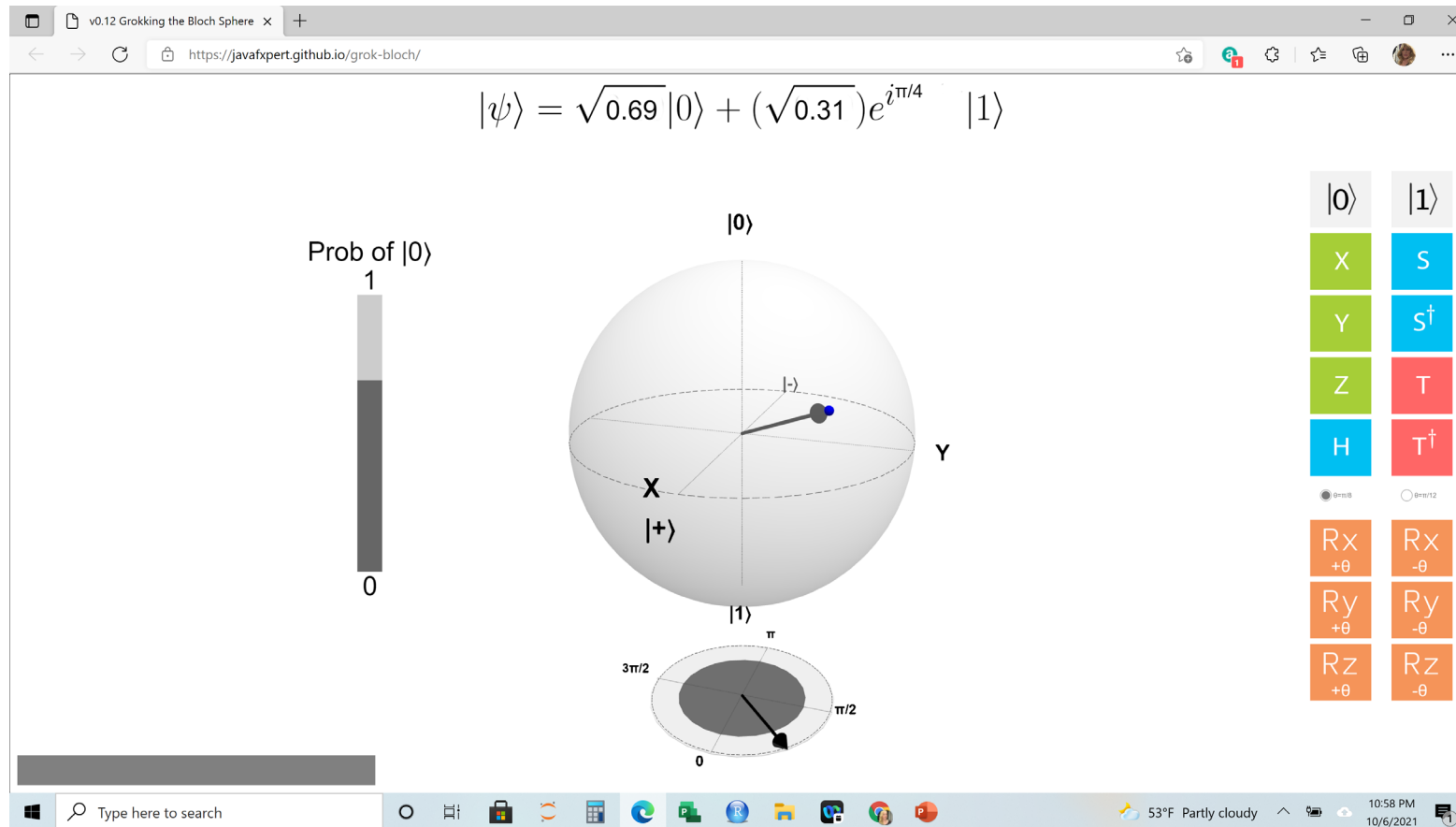
Quantum Computing

Example of Using Quantum Composer and learning about Quantum Gates

Nella Ludlow, PhD
Wright State University

Gronk the Bloch

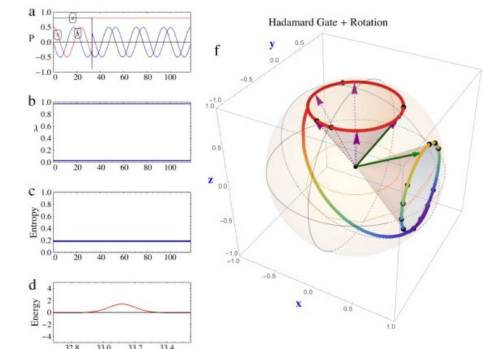
<https://javafxpert.github.io/grok-bloch/>



Professor Nella Ludlow - Wright State University

The Hadamard gate

While the Pauli gates are a lot like classic logic gates in some respects, the Hadamard gate, or *H-gate*, is a bona fide quantum beast. It shows up everywhere in quantum computing, and for good reason. The Hadamard gate has the characteristically *quantum* capacity to transform a definite quantum state, such as spin-up, into a murky one, such as a superposition of *both* spin-up and spin-down at the same time.



IBM Quantum Composer

- <https://quantum-computing.ibm.com/composer>

